

LION BUOY 61002, France

WMO: 996170

Lat: 42.100N

Lon: 4.700E

Elev: 3

StdP: 101.29

Time Zone: 1.00 (EUC)

Period: 02-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.4	5.9	-5.0	2.5	9.2	-2.1	3.2	10.9	22.1	7.3	21.1	8.5	17.1	350	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.1	26.0	23.7	25.4	23.0	24.7	22.3	24.2	25.5	23.6	25.0	23.0	24.4	4.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	18.6	25.2	23.2	17.9	24.7	22.5	17.2	24.2	72.9	25.6	70.7	25.1	68.2	24.5	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.8	18.3	16.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.3	11.5	10.4	12.2	13.9	16.1	19.7	22.3	22.7	21.1	18.4	14.9	12.3
	DBStd	4.64	1.99	2.50	1.58	1.44	1.46	1.85	1.72	1.49	1.94	2.51	2.41	2.09
	HDD10.0	46	11	23	4	0	0	0	0	0	0	0	1	7
	HDD18.3	1155	212	221	189	133	71	8	0	0	2	29	103	187
	CDD10.0	2359	57	35	73	117	189	291	381	395	332	262	149	78
	CDD18.3	426	0	0	0	0	2	48	122	137	84	32	1	0
	CDH23.3	615	0	0	0	0	0	12	237	283	78	5	0	0
	CDH26.7	3	0	0	0	0	0	0	2	1	0	0	0	0
Wind	WSAvg	7.7	8.5	9.4	8.6	6.7	7.3	6.2	6.6	6.7	7.0	7.7	8.9	9.5
Precipitation	PrecAvg	794	59	51	56	68	72	57	46	57	86	102	76	64
	PrecMax	1106	196	140	139	168	158	202	90	113	162	215	232	173
	PrecMin	593	1	9	3	22	22	9	15	20	26	38	11	8
	PrecStd	136	45	32	33	38	30	35	19	23	31	53	48	45
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.6	14.3	15.6	18.1	20.2	24.6	27.0	26.6	25.2	24.2	20.2	17.5
		MCWB	12.6	11.6	12.8	15.9	17.6	22.5	24.7	23.2	22.3	22.2	17.7	15.9
	2%	DB	14.7	13.7	14.9	17.3	19.3	23.5	26.3	26.0	24.7	22.9	19.3	16.2
		MCWB	12.1	11.1	12.3	15.2	17.2	21.6	24.2	23.4	21.9	20.7	17.2	14.5
	5%	DB	14.2	13.3	14.2	16.2	18.7	22.7	25.6	25.4	24.3	22.2	18.6	15.3
		MCWB	11.8	10.9	11.8	13.9	16.7	20.5	23.4	23.1	21.6	20.2	16.6	13.2
	10%	DB	13.8	13.0	13.8	15.7	18.1	22.1	24.6	24.9	23.8	21.7	18.0	14.7
		MCWB	11.5	10.7	11.5	13.2	16.1	20.0	22.3	22.7	21.2	19.6	16.1	12.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.7	13.2	13.7	16.5	18.7	23.1	25.7	24.6	23.6	23.4	18.6	16.0
		MCDB	14.5	13.7	14.9	17.6	19.6	23.8	26.5	25.6	24.6	24.0	19.3	17.1
	2%	WB	13.2	12.6	13.2	15.6	17.9	21.7	24.6	24.1	22.9	21.5	17.7	14.6
		MCDB	14.0	13.4	14.2	16.8	18.8	22.6	26.0	25.3	24.1	22.4	18.7	15.6
	5%	WB	12.7	12.2	12.8	14.7	17.3	21.1	23.8	23.7	22.3	20.9	17.0	13.6
		MCDB	13.7	13.2	13.8	15.9	18.5	22.1	25.4	25.0	23.8	22.1	18.2	14.6
	10%	WB	12.3	11.7	12.5	14.2	16.7	20.5	22.9	23.2	21.8	20.0	16.2	12.9
		MCDB	13.5	12.9	13.6	15.3	18.0	21.6	24.2	24.6	23.6	21.3	17.8	14.1
Mean Daily Temperature Range	5% DB	MDBR	2.1	2.3	1.9	1.8	1.8	1.9	1.9	2.1	2.2	2.1	1.9	2.0
		MCDBR	2.1	2.3	1.9	2.1	2.2	2.2	2.1	2.3	2.4	2.0	2.3	2.1
		MCWBR	2.4	2.5	1.9	1.8	2.0	2.3	2.1	2.3	2.7	2.4	2.8	2.7
	5% WB	MCDBR	1.8	2.0	1.7	2.0	2.1	2.0	1.7	2.0	2.3	1.8	2.1	1.9
		MCWBR	2.4	2.6	1.8	1.8	2.1	2.5	2.1	2.5	2.8	2.3	2.7	2.5
Clear-Sky Solar Irradiance	taub	0.332	0.348	0.387	0.410	0.418	0.419	0.419	0.415	0.402	0.391	0.359	0.334	
	taud	2.498	2.456	2.358	2.303	2.331	2.367	2.376	2.396	2.414	2.423	2.449	2.487	
	Ebn at Noon	812	855	856	860	859	857	853	846	831	792	767	774	
	Edh at Noon	74	89	110	125	125	121	118	113	104	92	77	69	
All-Sky Solar Radiation	RadAvg	1.91	2.82	4.21	5.55	6.60	7.51	7.36	6.46	4.87	3.26	2.13	1.68	
	RadStd	0.17	0.30	0.33	0.34	0.42	0.24	0.27	0.27	0.22	0.26	0.20	0.15	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (20 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+46	

Nomenclature: See separate page